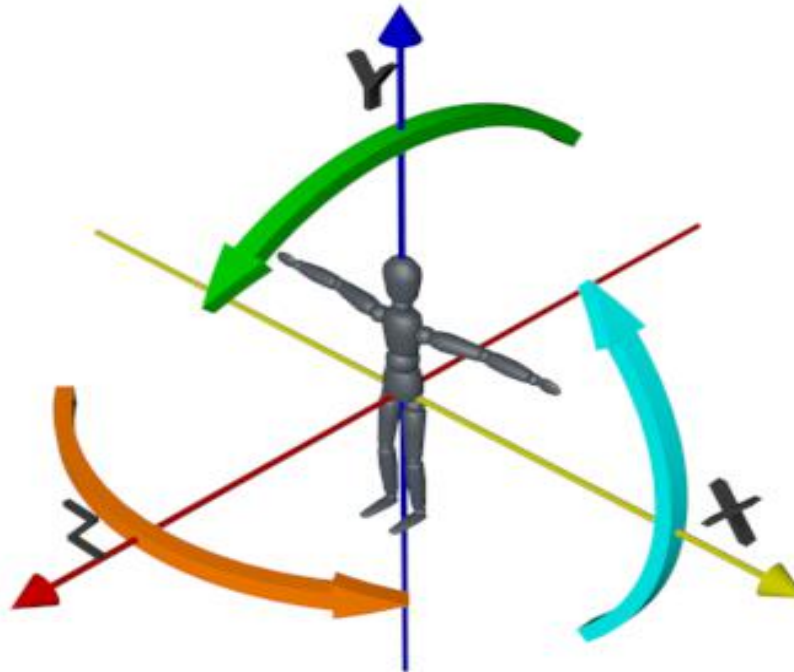




7.1 Rotational kinematics

FTF:8H, NTF:8H

What is a RIGID BODY?



HUMAN



DOOR



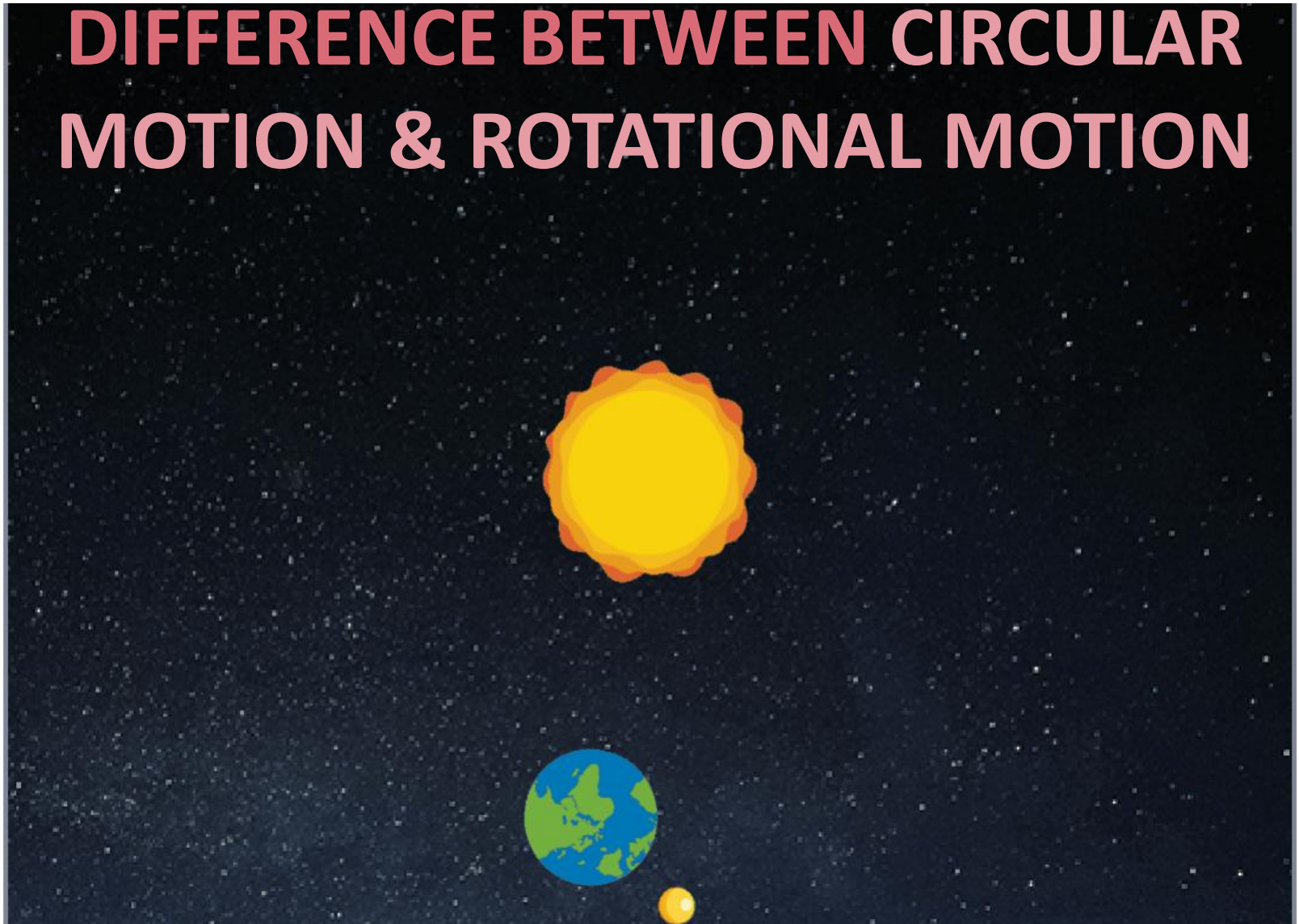
HELICOPTER

**What do these
3 objects have
in common?**



A **RIGID BODY** is an object which when a **FORCE** is applied on it, it won't change its size or shape

DIFFERENCE BETWEEN CIRCULAR MOTION & ROTATIONAL MOTION





Learning Outcomes

7.1 ROTATIONAL KINEMATICS

At the end of this chapter, students should be able to:

a) Define

- i. Angular Displacement, θ
- ii. Average Angular Velocity, ω_{av}
- iii. Instantaneous Angular Velocity, ω_{inst}
- iv. Average Angular Acceleration, α_{av}
- v. Instantaneous Angular Acceleration, α_{inst}

b) Use

- i. Angular Displacement, θ
- ii. Average Angular Velocity, ω_{av}
- iii. Instantaneous Angular Velocity, ω_{inst}
- iv. Average Angular Acceleration, α_{av}
- v. Instantaneous Angular Acceleration, α_{inst}



Learning Outcomes

7.1 ROTATIONAL KINEMATICS

At the end of this chapter, students should be able to:

- c)** State parameters in rotational motion with their corresponding quantities in linear motion
- d) Use** parameters in rotational motion with their corresponding quantities in linear motion

$$s = r\theta$$

$$v = r\omega$$

$$a_t = r\alpha$$

$$a_c = \frac{v^2}{r} = r\omega^2$$

 **Learning
Outcomes**

7.1 ROTATIONAL KINEMATICS

At the end of this chapter, students should be able to:

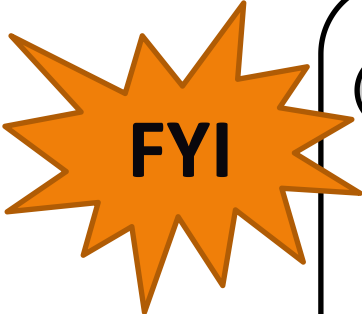
- e) Solve** problem related to rotational motion with constant angular acceleration

$$\omega = \omega_0 + \alpha t$$

$$\theta = \omega_0 t + \frac{1}{2} \alpha t^2$$

$$\omega^2 = \omega_0^2 + 2\alpha\theta$$

CONCEPT SIMILAR TO LINEAR KINEMATICS (CHAPTER 2)

**FYI**

Rotational Kinematics
is a mathematical
branch of physics that
**uses equations to
describe the motion of
object without
reference to FORCES**



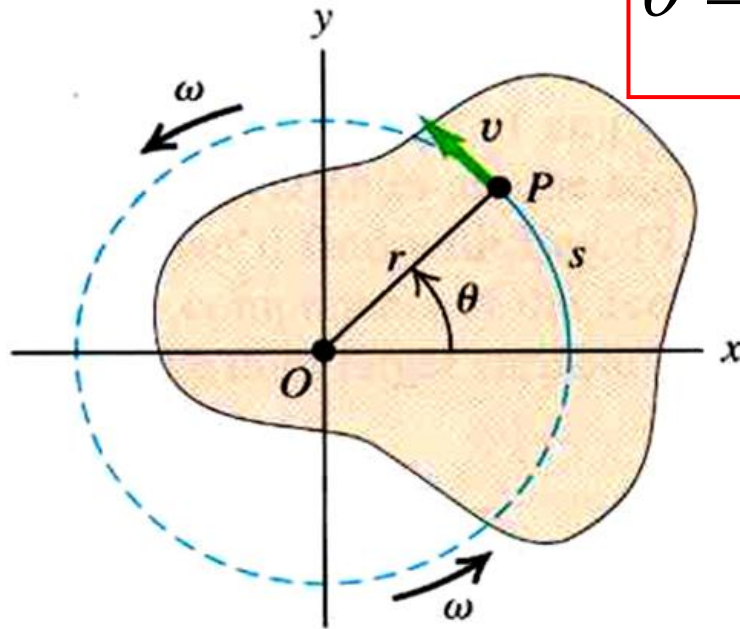
Angular Displacement (θ)

SIMILAR TO CHAP 6

- angle undergone by a particle from a fixed reference point.

$$\theta = \frac{s}{r}$$

where

 s = linear distance r = distance from rotational axis-- Unit SI: **radian (rad)**,-- another unit: revolution (rev)
(*1 rev = 2π rad)

Angular velocity (ω)

SIMILAR TO CHAP 6

- Angular velocity is the rate of change of angular displacement
- Unit SI : **rad s⁻¹**
- Others units used : rps (revolutions per second),
rpm (revolutions per minute)

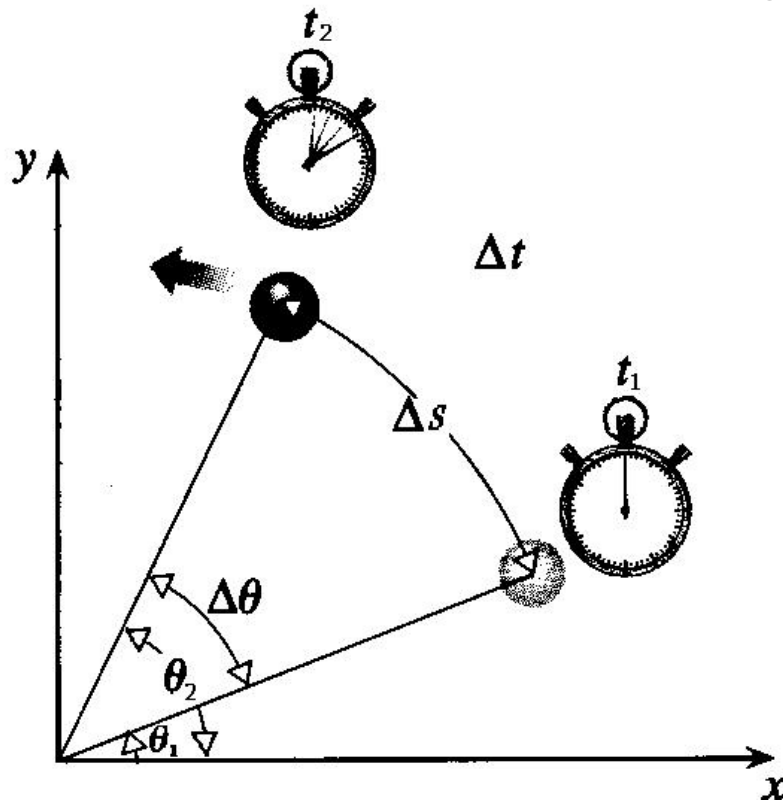
$$1 \text{ rpm} = \frac{2\pi}{60} \text{ rad s}^{-1} = \frac{\pi}{30} \text{ rad s}^{-1}$$

- Vector quantity

Sometimes also referred as
Angular Frequency

Average angular velocity (ω_{av})

$$\omega_{av} = \frac{\text{change in angular displacement}}{\text{time taken}}$$



$$\omega_{av} = \frac{\Delta\theta}{\Delta t} = \frac{\theta_2 - \theta_1}{t_2 - t_1}$$

where

θ_1 = initial angular displacement

θ_2 = final angular displacement

t_1 = initial time

t_2 = final time

Instantaneous angular velocity (ω_{inst})

angular velocity **at a specified position** or **at particular of time** (Δt approaches zero).

$$\omega = \lim_{\Delta t \rightarrow 0} \frac{\Delta \theta}{\Delta t}$$

$$\omega = \frac{d\theta}{dt}$$

-- commonly referred as
**angular velocity at
point A or angular
velocity at time t.**

Angular acceleration (α)

-- Rate of change of angular velocity of a rotating rigid body.

-- Unit SI : **rad s⁻²**

-- vector quantity

Average angular acceleration (α_{av})

$$\alpha_{av} = \frac{\text{Change in angular velocity}}{\text{time taken}}$$

$$\alpha_{av} = \frac{\omega_2 - \omega_1}{t_2 - t_1}$$

where

ω_1 = initial angular velocity

ω_2 = final angular velocity

t_1 = initial time

t_2 = final time

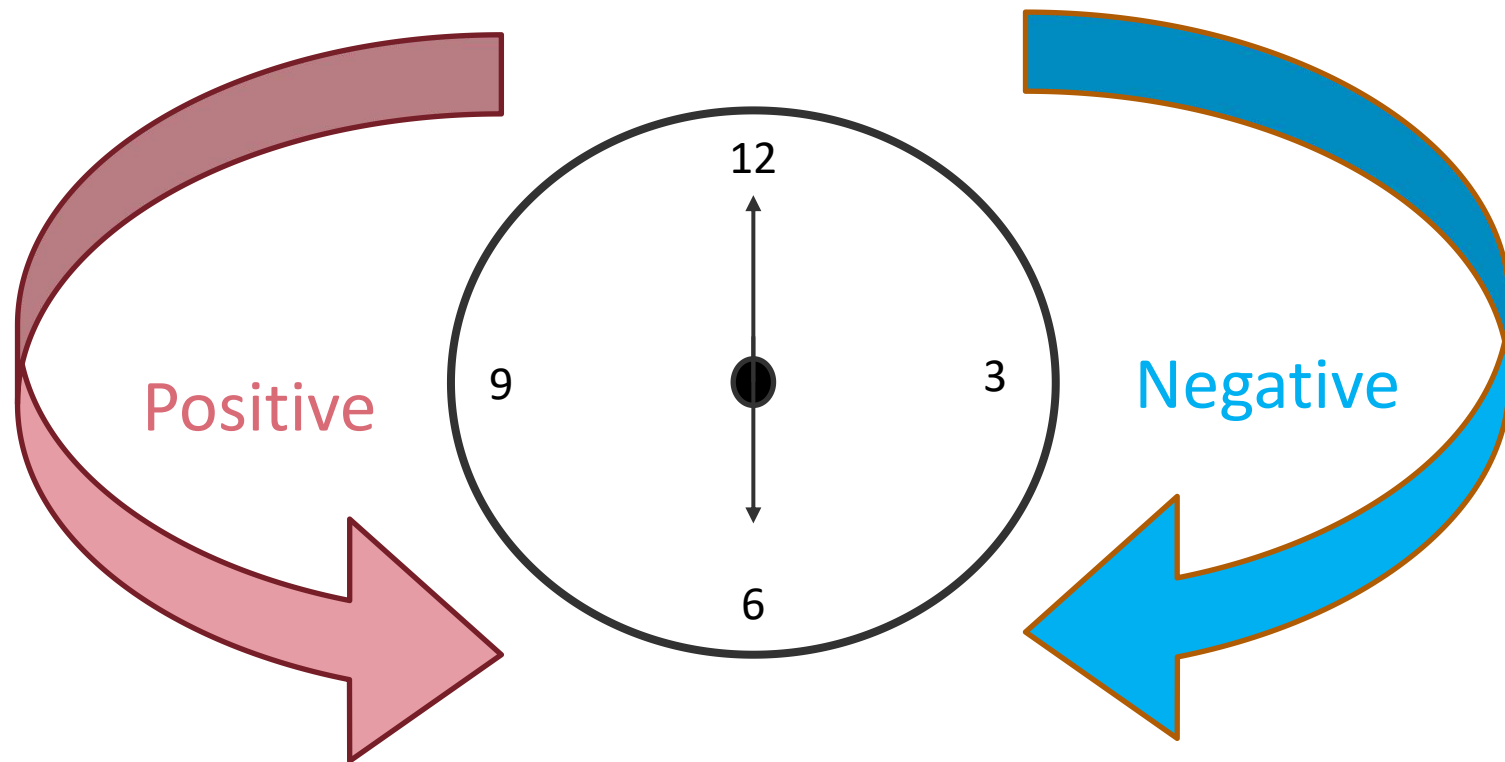
Instantaneous angular acceleration (α_{inst})

-- angular acceleration **at a specified position** or **instant of time** ($\Delta t \rightarrow \text{zero}$).

$$\alpha = \lim_{\Delta t \rightarrow 0} \frac{\Delta \omega}{\Delta t} = \frac{d\omega}{dt}$$

The value of θ , ω & α are always
CONSTANT (fixed reference axis)

- By sign convention, **positive (+)** if object rotates **counterclockwise** and **negative (-)** if it rotates **clockwise**.



Example 1

**Conversion between degrees, radian
& revolution or rotation.**

(1) $\pi / 9$ radian $\rightarrow$ degrees

(2) $60^\circ \rightarrow$ radian

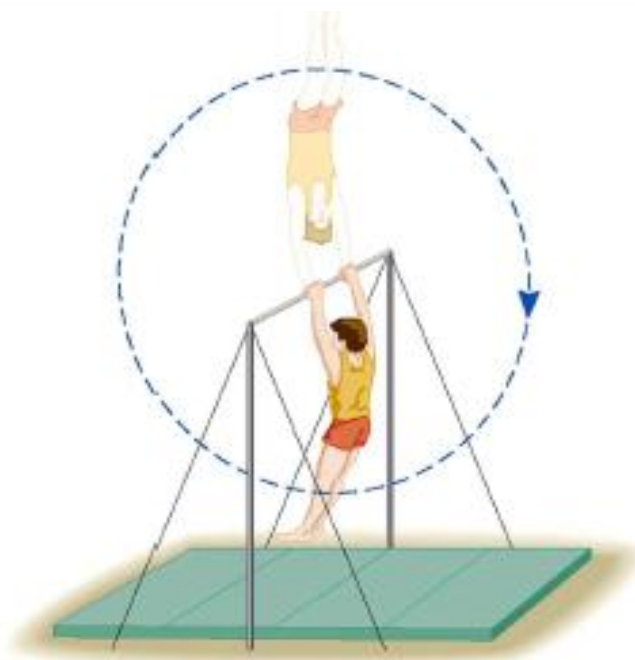
(3) 3 rev $\rightarrow$ rad

(4) 11 rad $\rightarrow$ rev

(5) 120 rpm $\rightarrow$ rad s⁻¹

(6) 50 rps $\rightarrow$ rad s⁻¹

7.



A gymnast on a high bar swings through two revolutions in a time of 1.90s. Find the average **angular velocity** (in rad/s) of the gymnast.

Answer : 6.61 rad/s

ANSWER

$$(1) \pi/9 = \frac{180}{9} = 20^\circ$$

$$(2) 60^\circ \times \frac{\pi}{180} = \frac{\pi}{3} \text{ rad}$$

$$(3) 3 \text{ rev} \times 2\pi = 6\pi \text{ rad}$$

$$(4) 11 \text{ rad} \div 2\pi = 1.75 \text{ rev}$$

$$(5) 120 \text{ rpm} \times \frac{2\pi}{60} = 12.57 \text{ rad s}^{-1}$$

$$(6) 50 \text{ rps} \times \frac{2\pi}{1} = 314.16 \text{ rad s}^{-1}$$

(7)

$$\omega_{av} = \frac{\Delta\theta}{\Delta t} = \frac{2(2\pi)}{1.90} = 6.61 \text{ rad s}^{-1}$$

Example 2

The instantaneous angular velocity, ω of the flywheel is given by

$$\omega = 8t^3 - t^2$$

where ω in radian per second and t in seconds.

Determine

- the average angular acceleration between 2.2 s and 4.8 s,
- the instantaneous angular acceleration at time, 3.0 s.

Solution :**Refer to:** $\omega = 8t^3 - t^2$ **a. At time, $t_1 = 2.2$ s :**

$$\omega_1 = 8(2.2)^3 - (2.2)^2$$

$$\omega_1 = 80.3 \text{ rad s}^{-1}$$

At time, $t_2 = 4.8$ s :

$$\omega_2 = 8(4.8)^3 - (4.8)^2$$

$$\omega_2 = 862 \text{ rad s}^{-1}$$

Therefore the average angular acceleration is

$$\alpha_{\text{av}} = \frac{\omega_2 - \omega_1}{t_2 - t_1} = \frac{862 - 80.3}{4.8 - 2.2}$$

$$\alpha_{\text{av}} = 301 \text{ rad s}^{-2}$$

Solution :**b. The instantaneous angular acceleration as a function of time is**

$$\alpha = \frac{d\omega}{dt}$$

$$= \frac{d}{dt}(8t^3 - t^2)$$

$$\alpha = 24t^2 - 2t$$

At time, $t = 3.0$ s :

$$\alpha = 24(3.0)^2 - 2(3.0)$$

$$\alpha = 210 \text{ rad s}^{-2}$$

Example 3

The angular displacement, θ of the wheel is given by

$$\theta = 5t^2 - t$$

where θ in radians and t in seconds. The diameter of the wheel is 0.56 m.

Determine

- (a) the **angle, θ in degree**, at time 2.2 s and 4.8 s,
- (b) the **distance** that a particle on the rim moves during that time interval,
- (c) the **average angular velocity**, in rad s^{-1} and in rev min^{-1} (rpm), between 2.2 s and 4.8 s,
- (d) the **instantaneous angular velocity** at time 3.0 s.

Solution : $r = \frac{d}{2} = \frac{0.56}{2} = 0.28 \text{ m}$

a. At time, $t_1 = 2.2 \text{ s}$:

Use $\theta = 5t^2 - t = 5(2.2)^2 - 2.2$

$$\theta_1 = 22 \text{ rad}$$

$$= 22 \left(\frac{180}{\pi} \right) = 1261^\circ$$

At time, $t_2 = 4.8 \text{ s}$:

Use $\theta = 5t^2 - t = 5(4.8)^2 - 4.8$

$$\theta_2 = 110 \text{ rad}$$

$$= 110 \left(\frac{180}{\pi} \right) = 6303^\circ$$

b. By applying the equation of arc length,
Therefore

$$s = r\theta$$

$$\Delta s = r\Delta\theta = r(\theta_2 - \theta_1)$$

$$= 0.28(110 - 22)$$

$$\Delta s = 24.6 \text{ m}$$

c. The average angular velocity in rad s^{-1} is given by

$$\omega_{\text{av}} = \frac{\Delta\theta}{\Delta t} = \frac{(\theta_2 - \theta_1)}{(t_2 - t_1)}$$

$$= \frac{(110 - 22)}{(4.8 - 2.2)}$$

$$\omega_{\text{av}} = 33.9 \text{ rad s}^{-1}$$

c. and the average angular velocity in rev min^{-1} is

$$\omega_{\text{av}} = \left(\frac{33.9 \text{ rad}}{1 \text{ s}} \right) \left(\frac{1 \text{ rev}}{2\pi \text{ rad}} \right) \left(\frac{60 \text{ s}}{1 \text{ min}} \right)$$
$$\omega_{\text{av}} = 324 \text{ rev min}^{-1} \text{ OR } 324 \text{ rpm}$$

d. The instantaneous angular velocity as a function of time is

$$\omega = \frac{d\theta}{dt}$$
$$= \frac{d}{dt}(5t^2 - t)$$
$$\omega = 10t - 1$$

At time, $t = 3.0 \text{ s}$:

$$\omega = 10(3.0) - 1$$
$$\omega = 29 \text{ rad s}^{-1}$$



Learning Outcomes

7.1 ROTATIONAL KINEMATICS

At the end of this chapter, students should be able to:

- c)** State parameters in rotational motion with their corresponding quantities in linear motion
- d) Use** parameters in rotational motion with their corresponding quantities in linear motion

$$s = r\theta$$

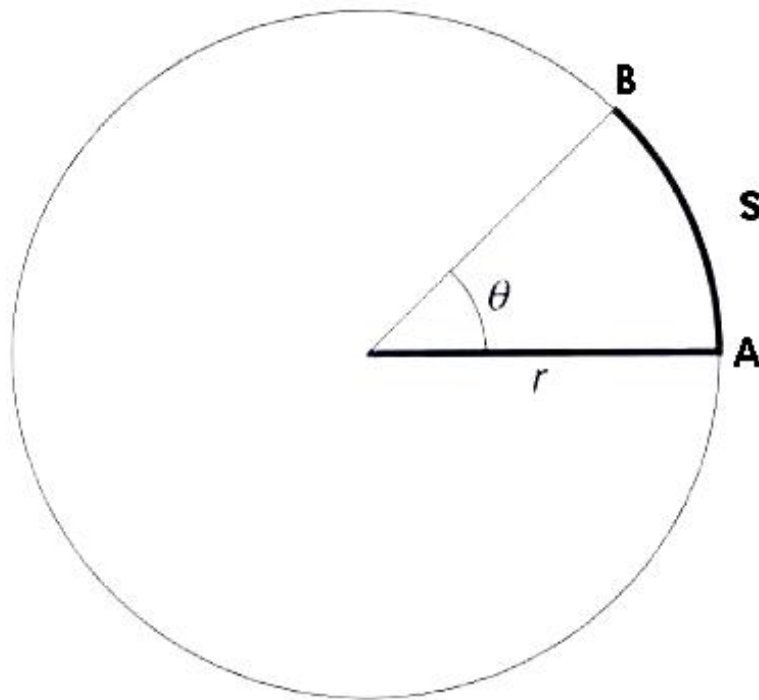
$$v = r\omega$$

$$a_t = r\alpha$$

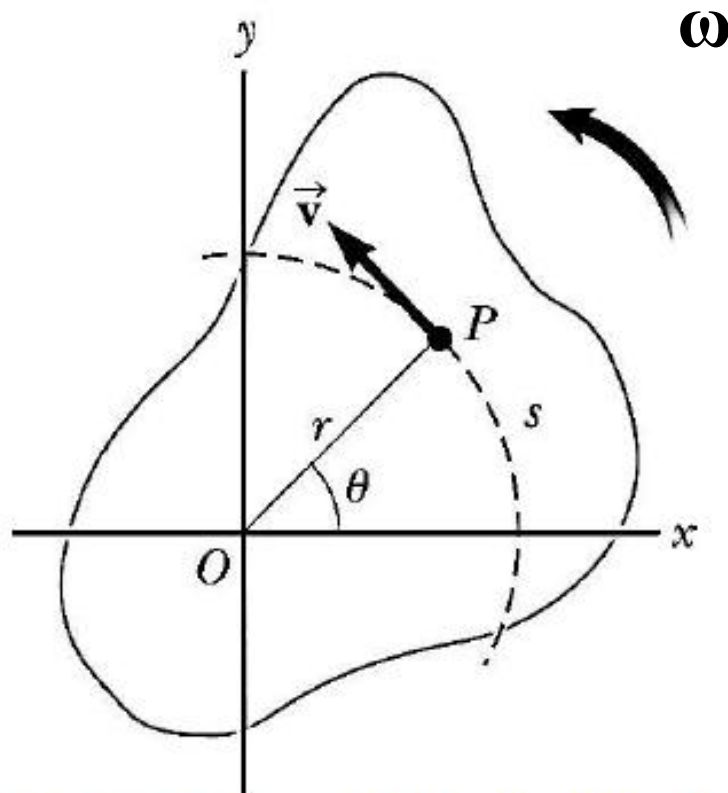
$$a_c = \frac{v^2}{r} = r\omega^2$$

Linear and Rotational Quantities (LO 7.1 c)

Quantity	Linear	Rotational	Relation
Displacement	s	θ	$s = r\theta$
Velocity	v	ω	$v = r\omega$
Acceleration	a_t	α	$a_t = r\alpha$



$$s = r\theta$$



$$v = r\omega$$

Figure 7-3 As a rigid body rotates about fixed axis through point O, the point P has a tangential velocity $\vec{v}$ that is always tangent to the circular path of radius r.

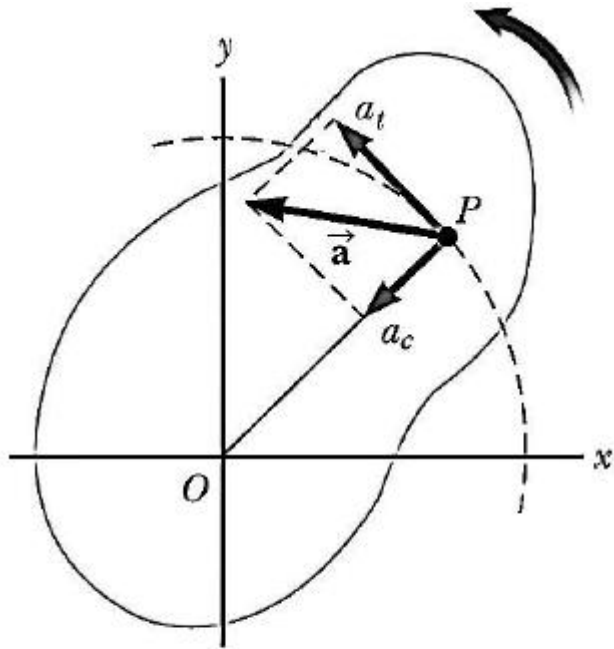


Figure 7-4 As a rigid body rotates about a fixed axis through point O, the point P experiences tangential (linear) acceleration a_t , and centripetal (linear) acceleration a_c .

1. Linear or Tangential acceleration, a_t
(in the direction of motion)

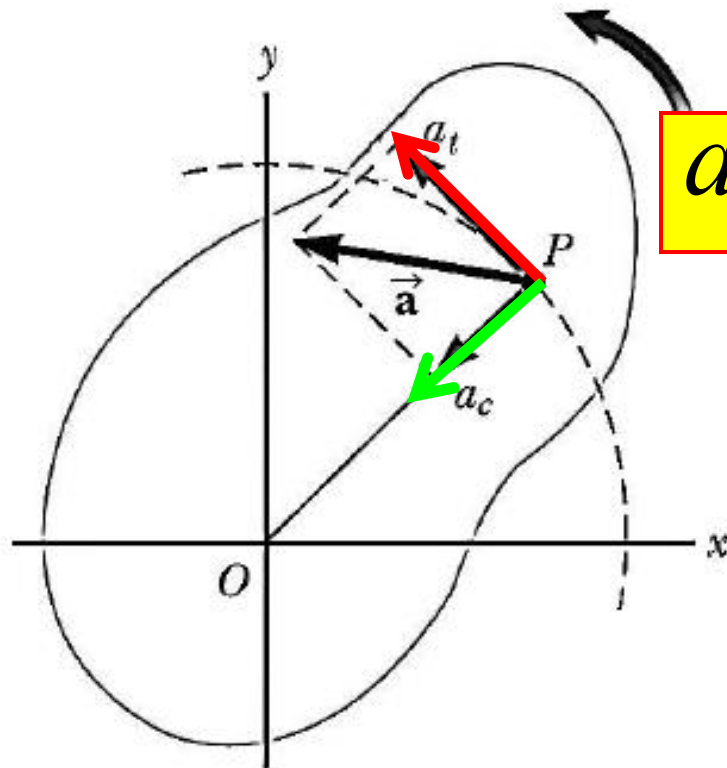
from $v = r\omega$

$$\frac{dv}{dt} = r \frac{d\omega}{dt} \quad (r \text{ is constant})$$

$$a_t = r\alpha$$

2. Centripetal acceleration, a_c
(towards the center)

$$a_c = \frac{v^2}{r} = r\omega^2$$



$$a_t = r\alpha$$

$$a_c = \frac{v^2}{r} = r\omega^2$$

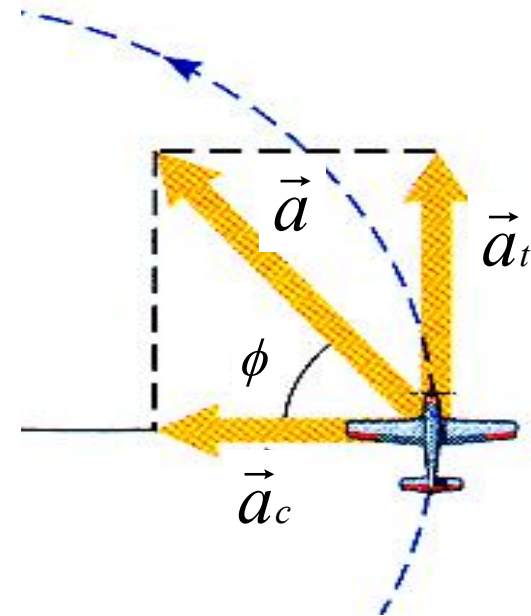
Figure 7-4 As a rigid body rotates about a fixed axis through point O, the point P experiences tangential (linear) acceleration a_t , and centripetal (linear) acceleration a_c .

Magnitude of total acceleration:

$$a = \sqrt{a_t^2 + a_c^2}$$

at an angle :

$$\tan \phi = \frac{a_t}{a_c}$$



Example 4

You and a friend are playing on the merry-go-round. You stand at the outer edge of the merry-go-round and your friend stands halfway between the outer edge and the center. Assume the rotation rate of the merry-go-round is constant.

Who has the greatest angular velocity?

1. You do
2. Your friend does
3. Same ← **CORRECT**

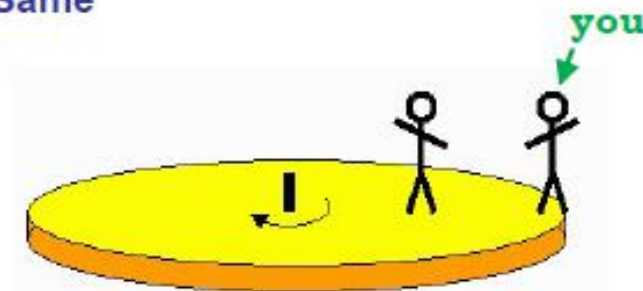


every point on a rigid body have the same value of θ , ω and α

Example 5

Who has the greatest tangential velocity?

1. You do ← **CORRECT**
2. Your friend does
3. Same

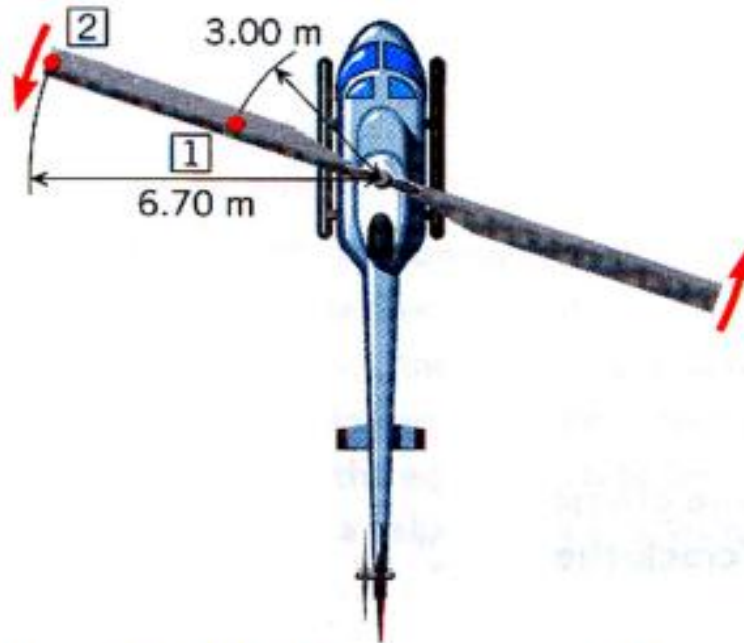


$v = r\omega$ (ω is the same but r is larger for you)

Who has the greatest centripetal acceleration?

1. You do ← **CORRECT**
2. Your friend does
3. Same

Centripetal acceleration is $a_c = r\omega^2$ and you have the largest radius.

Example 6

A helicopter blade has an angular velocity of $\omega = 6.50 \text{ rev s}^{-1}$ and an angular acceleration of $\alpha = 1.30 \text{ rev s}^{-2}$. For points 1 and 2 on the blade in figure above, find the magnitudes of (a) the **tangential velocities** and (b) the **tangential accelerations**.

Solution

$$\omega = \frac{6.5(2\pi) \text{ rad}}{1\text{s}} = 40.8 \text{ rad s}^{-1}$$

$$\alpha = \frac{1.30(2\pi) \text{ rad}}{1\text{s}^2} = 8.17 \text{ rad s}^{-2}$$

For a point 1 : $r = 3 \text{ m}$

$$v = r\omega = 3(40.8) = 122 \text{ m s}^{-1}$$

$$a_t = r\alpha = 3(8.17) = 24.5 \text{ m s}^{-2}$$

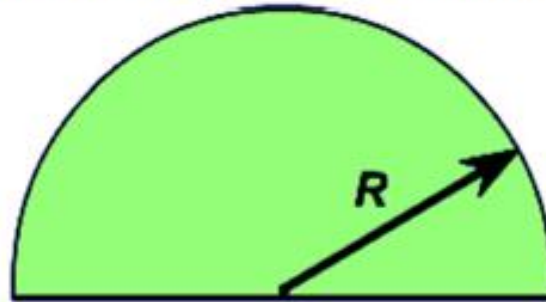
For a point 2 : $r = 6.7 \text{ m}$

$$v = r\omega = 6.70(40.8) = 273 \text{ m s}^{-1}$$

$$a_t = r\alpha = 6.70(8.17) = 54.7 \text{ m s}^{-2}$$

Exercise

- (1) A circle of radius $R = 10 \text{ cm}$ rotates with angular velocity $\omega = 680 \text{ rad s}^{-1}$. What is the time T for a full turn?

 $T =$

- ☐ 0.01 s
- ☐ 0.09 s
- ☐ 0.1 s
- ☐ 1 s

Determine the value of the linear velocity of points on the circumference of the circle.

 $v =$

Exercise

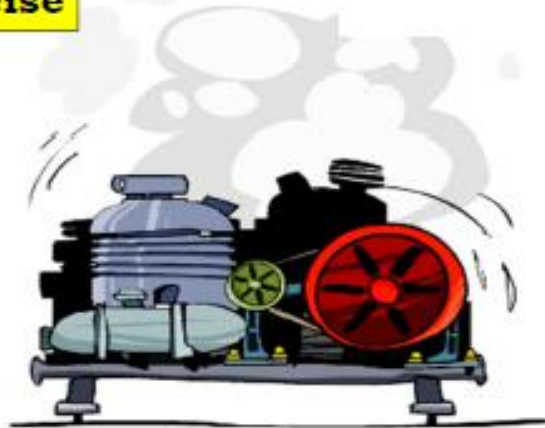
- (2) Calculate the average linear velocity of the points located on the Earth's equator and pole.
Assume $R = 6400 \text{ km}$.



Match the correct answers.

$$V_{eq} \approx \bullet \quad \bullet \quad 0 \text{ m s}^{-1}$$

$$V_{pol} \approx \bullet \quad \bullet \quad 465 \text{ m s}^{-1}$$

Exercise

- (3) A flywheel makes 180 revolutions per minute. Due to friction in the bearings it stops after two minutes. Calculate the average angular deceleration α .

$\alpha =$

- ☐ -0.471 rad s⁻²
- ☐ -0.157 rad s⁻²
- ☐ 0.157 rad s⁻²
- ☐ 0.942 rad s⁻²

Exercise

- (4) Calculate the radius R of a flywheel, if the linear velocity of points on its circumference $v_1 = 6 \text{ m s}^{-1}$ and the points located closer to the axis by $l = 15 \text{ cm}$ move with the linear velocity $v_2 = 5.5 \text{ m s}^{-1}$.



- ☐ 0.018 cm
- ☐ 0.18 m
- ☐ 1.8 m
- ☐ 18 m

ANSWER**(1)**

$$T = \frac{2\pi}{\omega}$$

$$T = 0.01 \text{ s}$$

The answer is:
A. $T = 0.01 \text{ s}$

$$v = \omega R$$

$$v = 68 \text{ m s}^{-1}$$

The answer is:
B. $v = 68 \text{ m s}^{-1}$

(2)

$$v = \omega r = \frac{2\pi}{T} r$$

$$T = 86400 \text{ s}$$

on the equator $r = R$; $v = \frac{2\pi}{T} R = 465 \text{ m s}^{-1}$

on the pole $r = 0$; $v_{\text{pol}} \approx 0$

The answers:

$$v_{\text{eq}} \approx \quad 0 \text{ m s}^{-1}$$

$$v_{\text{pol}} \approx \quad 465 \text{ m s}^{-1}$$

(3)

$$\alpha = \frac{\Delta\omega}{t} = \frac{\omega_i - \omega_f}{t} = \frac{\omega_i}{t} = \frac{2\pi n}{t}$$

$$n = 180 \text{ rev min}^{-1} = 3 \text{ rev s}^{-1}, t = 2 \text{ min} = 120 \text{ s}$$

$$\alpha = 0.157 \text{ rad s}^{-2}$$

The answer is:

$$\text{B. } \alpha = 0.157 \text{ rad s}^{-2}$$

(4)

$$v_1 = \omega R$$

$$v_2 = \omega(R - \ell)$$

$$\frac{v_2}{v_1} = \frac{R - \ell}{R} \Rightarrow Rv_2 = Rv_1 - \ell v_1$$

$$R = \frac{\ell v_1}{v_1 - v_2}$$

$$R = 1.8 \text{ m}$$



Learning Outcomes

7.1 ROTATIONAL KINEMATICS

At the end of this chapter, students should be able to:

- e) Solve** problem related to rotational motion with constant angular acceleration

$$\omega = \omega_0 + \alpha t$$

$$\theta = \omega_0 t + \frac{1}{2} \alpha t^2$$

$$\omega^2 = \omega_0^2 + 2\alpha\theta$$

CONCEPT SIMILAR TO LINEAR KINEMATICS (CHAPTER 2)

Rotational motion with uniform (constant) angular acceleration.

When the angular velocity of rigid body increases uniformly from ω_0 to ω in time t , then: Uniform angular acceleration is:

$$\alpha = \frac{\omega - \omega_0}{t}$$

$$\omega = \omega_0 + \alpha t \quad \dots (1)$$

The angular displacement in time t :

$$\theta = (\text{Average angular velocity})(\text{time})$$

$$\theta = \frac{1}{2}(\omega_0 + \omega)t \quad \dots (2)$$

Substitute (1) into (2) :

$$\theta = \frac{1}{2}(\omega_0 + \omega_0 + \alpha t)t$$

$$\theta = \omega_0 t + \frac{1}{2}\alpha t^2 \quad \dots (3)$$

Squaring equation (1):

$$\begin{aligned}\omega^2 &= (\omega_0 + \alpha t)^2 \\ &= \omega_0^2 + 2\omega_0\alpha t + \alpha^2 t^2 \\ &= \omega_0^2 + 2\alpha\left(\omega_0 t + \frac{1}{2}\alpha t^2\right)\end{aligned}$$

$$\omega^2 = \omega_0^2 + 2\alpha\theta \quad \dots (4)$$

Note:

if the rotation **slow down** $\rightarrow -\alpha$

if the rotation **speed up** $\rightarrow +\alpha$

Table below shows the comparison between linear and rotational kinematics.

Linear Motion	Rotational Motion
$v = u + a t$	$\omega = \omega_0 + \alpha t$
$s = \frac{1}{2} (u + v) t$	$\theta = \frac{1}{2} (\omega_0 + \omega) t$
$s = u t + \frac{1}{2} a t^2$	$\theta = \omega_0 t + \frac{1}{2} \alpha t^2$
$v^2 = u^2 + 2as$	$\omega^2 = \omega_0^2 + 2 \alpha \theta$

ω_0 : initial angular velocity

ω : final angular velocity

θ : angular displacement OR angle undergone

α : angular acceleration

t : time

Rotation	Linear
ω_0	u
ω	v
α	a
θ	s

Example 7

A wheel rotates with a constant angular acceleration of 2.5 rad s^{-2} . At time $t = 0$, the angular velocity of the wheel is 3.0 rad s^{-1} . Determine :

- (a) The angle turned through by the wheel in the first 2 seconds.
- (b) angular velocity at $t = 2 \text{ s}$.

Solution :

$$\alpha = 2.5 \text{ rad s}^{-2}$$

$$t = 0 \rightarrow \omega_0 = 3 \text{ rad s}^{-1}$$

$$\text{when } t = 2 \text{ s, } \theta = ? , \omega = ?$$

$$(a) \text{ Using: } \theta = \omega_0 t + \frac{1}{2} \alpha t^2$$

$$\theta = (3)(2) + \frac{1}{2}(2.5)(2)^2$$

$$\boxed{\theta = 11 \text{ rad}}$$

$$(b) \text{ Using: } \omega = \omega_0 + \alpha t$$

$$\omega = (3) + (2.5)(2)$$

$$\boxed{\omega = 8 \text{ rad s}^{-1}}$$

Example 8

A particle travels in a horizontal circle of radius 50 cm with angular velocity 120 rpm. Its angular velocity increases at a constant rate to 180 rpm in 10 s. At the 10th second determine :

- the angular acceleration
- the tangential linear acceleration.

Solution:

$$\overset{\omega_0}{120 \text{ rpm}} \xrightarrow{t = 10 \text{ s}} \overset{\omega}{180 \text{ rpm}}$$

$$\omega_0 = 120 \text{ rpm} = \frac{120(2\pi)}{60} = 4\pi \text{ rad s}^{-1}$$

$$\omega = 180 \text{ rpm} = \frac{180(2\pi)}{60} = 6\pi \text{ rad s}^{-1}$$

when $t = 10 \text{ s}$:

find : $\alpha = ?$; $a_T = ?$

Using : $\omega = \omega_0 + \alpha t$

$$6\pi = 4\pi + \alpha(10)$$

$$\boxed{\alpha = 0.628 \text{ rad s}^{-2}}$$

(b) from : $a_T = r\alpha$

$$= (50 \times 10^{-2}) 0.628$$

$$\boxed{= 0.314 \text{ m s}^{-2}}$$

Follow Up Exercise

1. The shaft of a pump starts from rest and has an angular acceleration of 3.0 rad s^{-2} for 18.0 s . At the end of this interval, what is (a) the shaft's angular velocity (b) the angle through which the shaft has turned ?

{ Ans: (a) 54 ; (b) 486 }

2. A flywheel has a constant angular deceleration of 2.0 rad s^{-2} .
 - (a) Find the angle through which the flywheel turns as it comes to rest from an angular velocity of 220 rad s^{-1} .
 - (b) Find the time required for the flywheel to come to rest.

{ Ans: (a) 12100 ; (b) 110 }

- (3) The blades of an electric blender are whirling with an **angular velocity** of $+375 \text{ rad/s}$ while the “puree” button is pushed in. When the “blend” button is pressed, the blades accelerate and reach a greater **angular velocity** after the blades have rotated through an **angular displacement** of $+44.0 \text{ rad}$ (seven revolutions). The **angular acceleration** has a constant value of $+1740 \text{ rad/s}^2$. Find the final angular velocity of the blades.



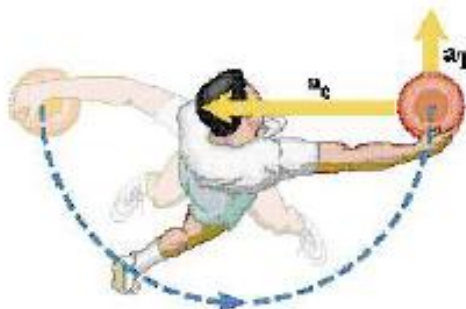
Answer : 542 rad/s

- (4) The blades of a ceiling fan start from rest and, after two revolutions, have an angular speed of 0.50 rev/s . The **angular acceleration** of the blades is constant. What is the angular speed after eight revolutions?

Answer : 1 rev/s

- (5) The blade of a lawn mower is rotating at an angular speed of 17 rev/s . The **tangential** speed of the outer edge of the blade is 32 m/s . What is the radius of the blade?

Answer : 0.3 m



- (6) Discus throwers often warm up by standing with both feet flat on the ground and throwing the discus with a twisting motion of their bodies. A top view of such a warm-up throw. Starting from rest, the thrower accelerates the discus to a final **angular velocity** of $+15.0$ rad/s in a time of 0.270 s before releasing it. During the acceleration, the discus moves on a circular arc of radius 0.810 m.
- Find (a) the magnitude a of the total **acceleration** of the discus just before it is released and (b) the angle ϕ that the total acceleration makes with the radius at this moment.

Answer for Follow Up Exercise

1. Given :

(a) $\alpha = 3\text{rads}^{-2}, t = 18\text{s}, \omega_0 = 0\text{rads}^{-1}$

$$\omega = \omega_0 + \alpha t$$

$$\omega = 3(18)$$

$$\omega = 54\text{rads}^{-1}$$

(b)

$$\theta = \omega_0 t + \frac{1}{2} \alpha t^2$$

$$\theta = 0(18) + \frac{1}{2} (3)(18)^2$$

$$\theta = 486\text{rad}$$

2. Given :

$$\alpha = -2 \text{ rad s}^{-2}, \omega = 0 \text{ rad s}^{-1},$$

$$\omega_0 = 220 \text{ rad s}^{-1}$$

(a)

$$\omega^2 = \omega_0^2 + 2\alpha\theta$$

$$0 = (-220)^2 + 2(-2)\theta$$

$$\theta = 12100 \text{ rad}$$

(b) $\omega = \omega_0 + \alpha t$

$$0 = (220) + (-2)t$$

$$t = 110 \text{ s}$$

3. Given :

$$\omega_o = 375 \text{ rads}^{-1}, \theta = 44 \text{ rad},$$

$$\alpha = 1740 \text{ rads}^{-2}$$

$$\omega^2 = \omega_o^2 + 2\alpha\theta$$

$$\omega^2 = (375)^2 + 2(1740)(44)$$

$$\omega = 542 \text{ rads}^{-1}$$

$$(4) \quad \omega_0 = 0 \text{ rad s}^{-1}$$

$$\theta = 2 \text{ rev}, \quad \omega = 0.5 \text{ rev/s} = \pi \text{ rad s}^{-1}$$

$$\theta = 8 \text{ rev}, \quad \omega = ?$$

$$\omega^2 = \omega_0^2 + 2\alpha\theta$$

$$(\pi)^2 = (0)^2 + 2\alpha(4\pi)$$

$$\alpha = \frac{\pi}{8} \text{ rad s}^{-2}$$

$$\omega^2 = \omega_0^2 + 2\alpha\theta$$

$$\omega^2 = (0)^2 + 2\left(\frac{\pi}{8}\right)(16\pi)$$

$$\omega = 2\pi \text{ rad s}^{-1}$$

$$\omega = 1 \text{ rev s}^{-1}$$

$$(5) \quad \omega = 17 \text{ rev s}^{-1} = 34\pi \text{ rad s}^{-1}$$

$$v = r\omega$$

$$32 = r(34\pi)$$

$$r = 0.3 \text{ m}$$

$$(6) \quad \omega = \omega_0 + \alpha t$$

$$15 = 0 + \alpha(0.27)$$

$$\alpha = 55.56 \text{ rad/s}$$

$$a_t = r\alpha$$

$$= 0.81(55.56) = 45 \text{ m/s}$$

$$a_c = r\omega^2$$

$$= 0.81(15)^2 = 182.25 \text{ m/s}^2$$

$$a = \sqrt{a_t^2 + a_c^2}$$

$$\tan \varphi = \frac{a_t}{a_c}$$

$$a = 182.25 \text{ m/s}^2$$

$$\Rightarrow \varphi = 13.48^\circ$$



End of Chapter 7